

Lab 09 – 18-11-2024

Task 1: Using **Doubly Header Circular Linked List** class **DQueue** implement following methods:

- a. `size()`:
 - To return number of elements
Elements: **3 4 5 1 7 9**
Size: 6
- b. `front()`:
 - To return the element from front
Elements: **3 4 5 1 7 9**
Front: 3
- c. `back()`:
 - To return the element from rear
Elements: **3 4 5 1 7 9**
Back: 9
- d. `push_front()`:
 - To Add an element to the front of the queue.
Elements: **3 4 5 1 7 9**
New: 10
New Elements: **10 3 4 5 1 7 9**
- e. `push_back()`:
 - To Add an element to the rear of the queue.
Elements: **3 4 5 1 7 9**
New: 10
New Elements: **3 4 5 1 7 9 10**
- f. `pop_front()`:
 - Remove and return the front element.
Elements: **3 4 5 1 7 9**
After: 4 5 1 7 9
- g. `pop_back()`:
 - Remove and return the rear element.
Elements: **3 4 5 1 7 9**
After: 3 4 5 1 7
- h. `is_empty()`:
 - To Return true if the queue is empty, otherwise false.
Elements: **3 4 5 1 7 9**
Empty: False
- i. `Clear()`:
 - Remove all elements from the queue.
Elements: **3 4 5 1 7 9**
After: No Elements

NOTE: Handle all the edge cases e.g. empty list and show appropriate message where applicable. Do Handle Memory leakage and dangling pointers also.